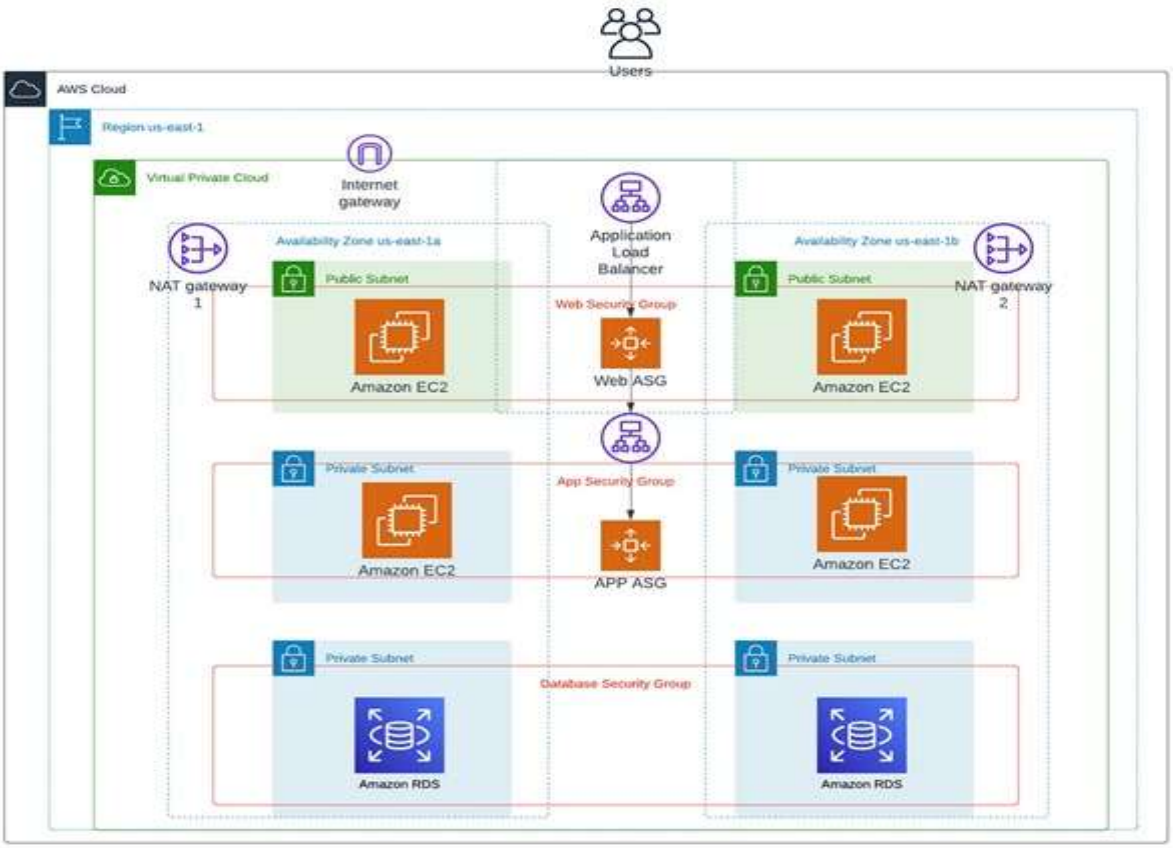
Project - 1
3 - Tier Architecture on AWS



Architecture overview:

3 -tier architecture which the application is divided into three layers. The presentation layer,application layer,and database layer .this separation makes systems more scalable ,secure and easier to maintain.

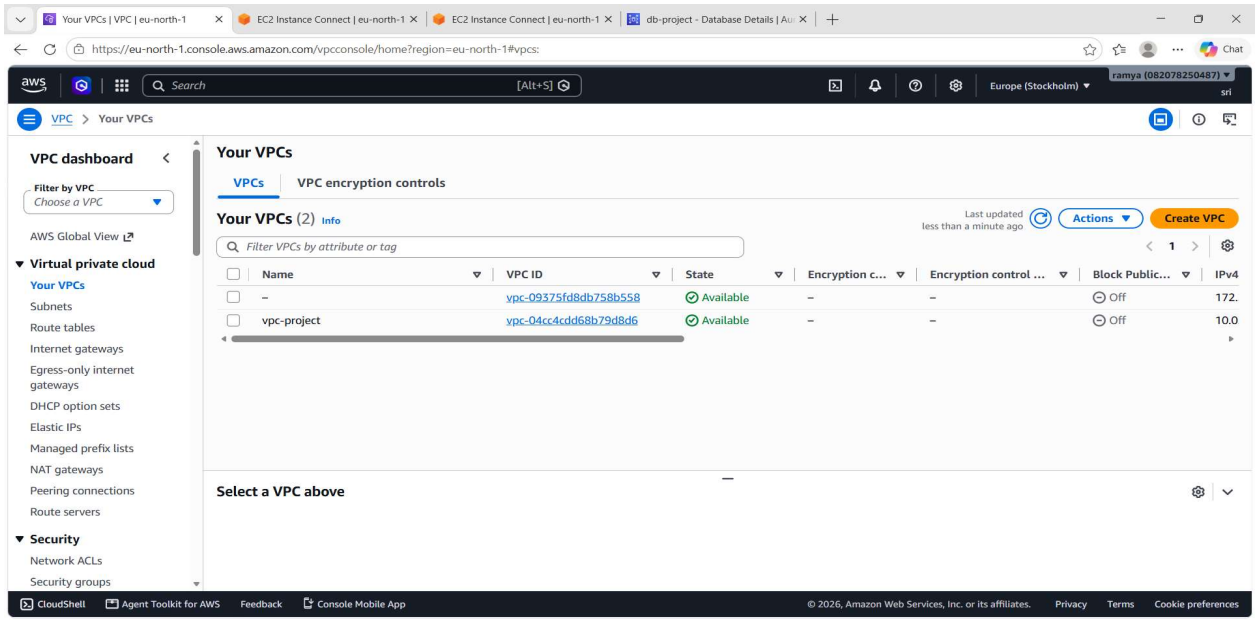
1. **Presentation Layer:** user interface where user interact with the system.
2. **Application layer:** process request ,applies the business rules,coordinate between UI and database.
3. **Data layer:** stores and manages application data

Steps to create the 3 - tier architecture

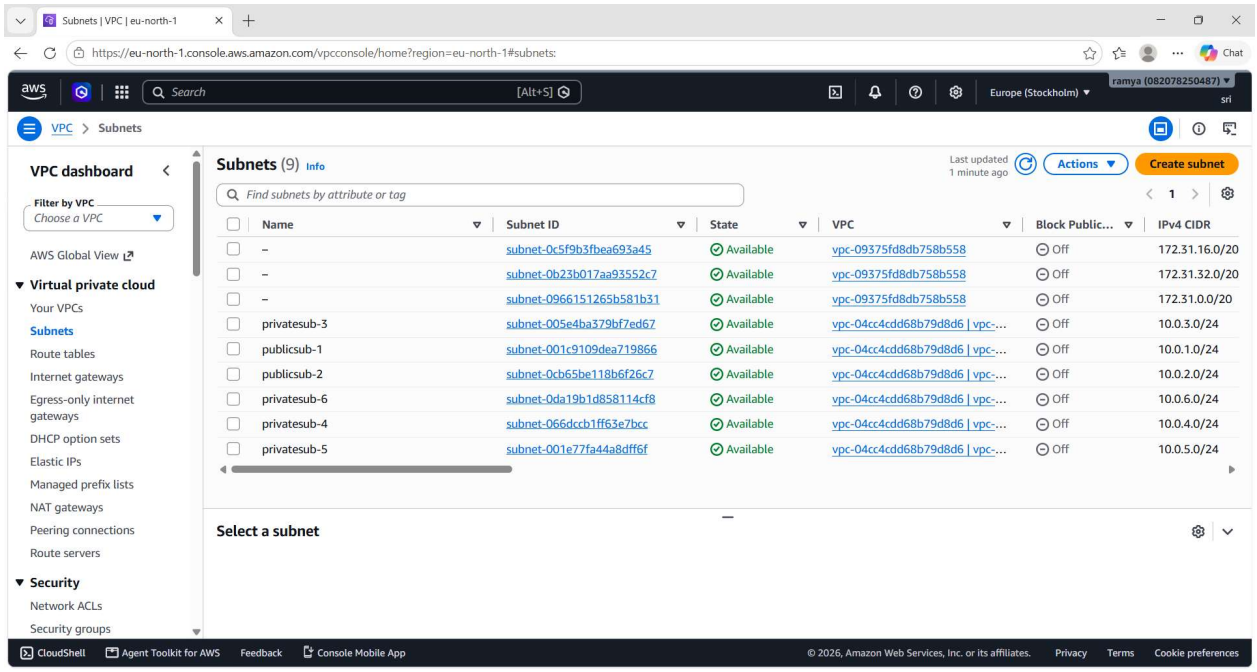
1. Create the Vpc ,subnets - 6,routetable -2,internet gateway - 1,NAT gateway.
2. Launch the EC2 instances.
3. Create the load balancer
4. Create the AMI
5. Create the auto scaling group,create launch template
6. Create the subnet group
7. Create the RDS(Database)
8. Establish the connection

Step-1: Create the VPC and its components

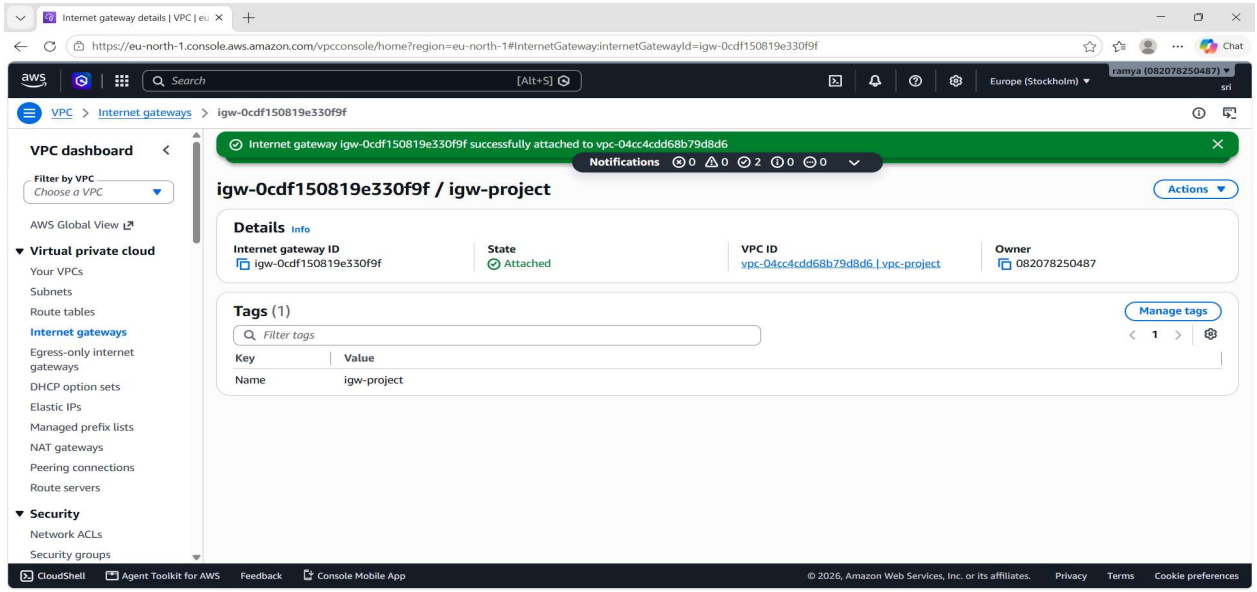
1.create the vpc



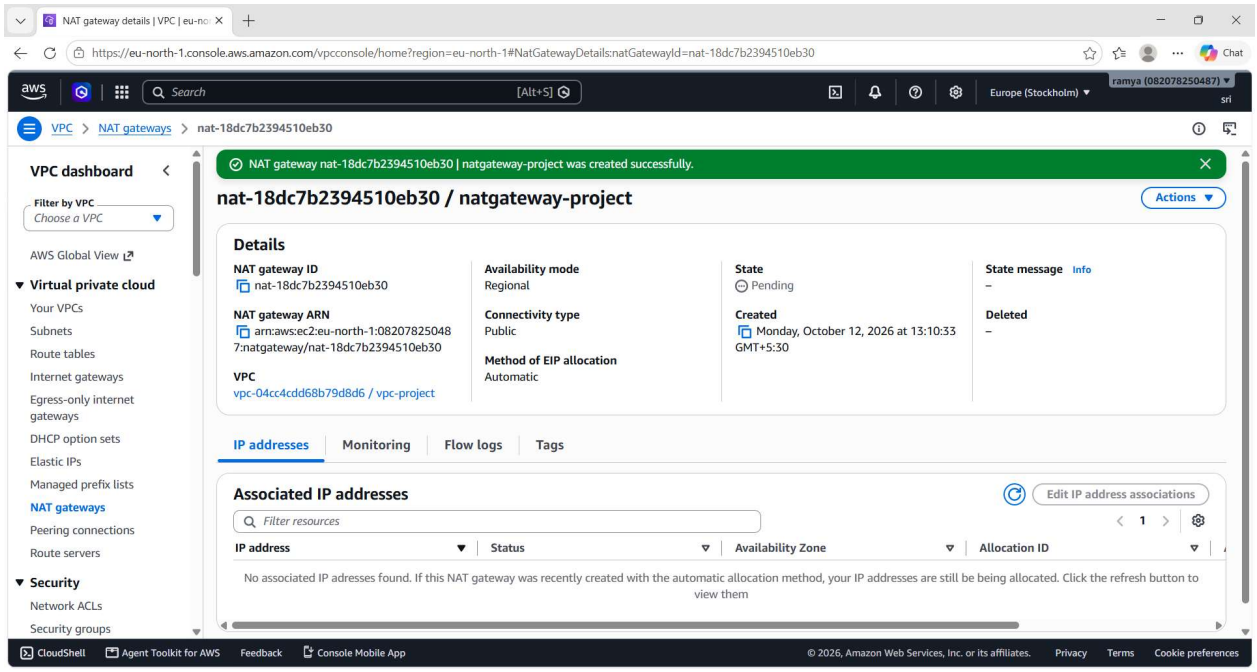
2. Create the subnets - 2 public and 4 private subnets



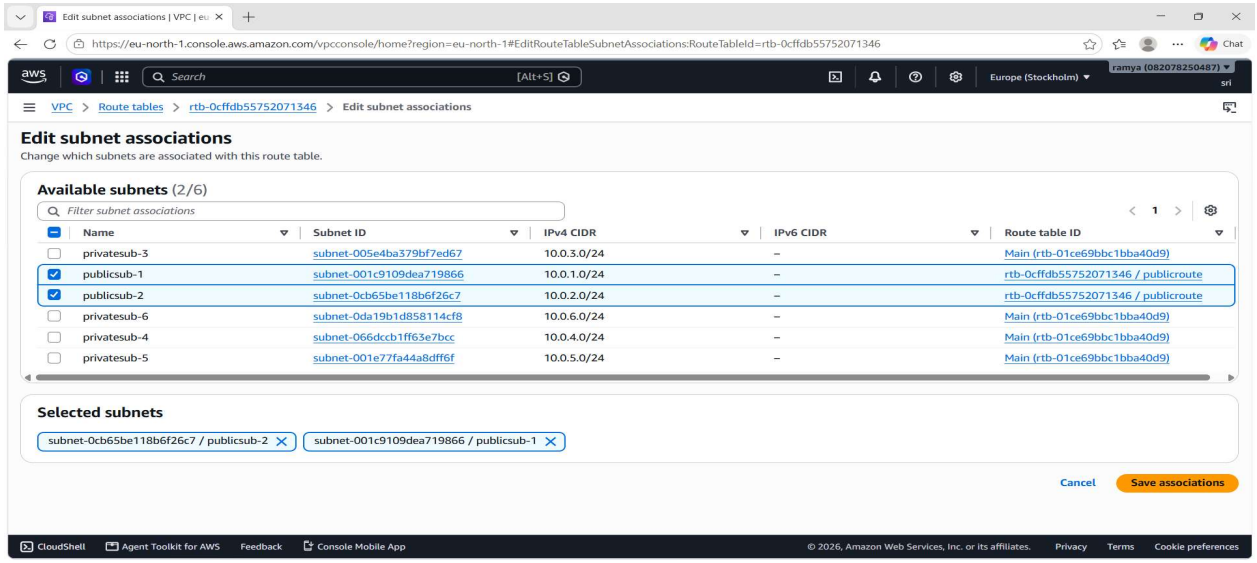
3. Create the internet gateway and attach to vpc

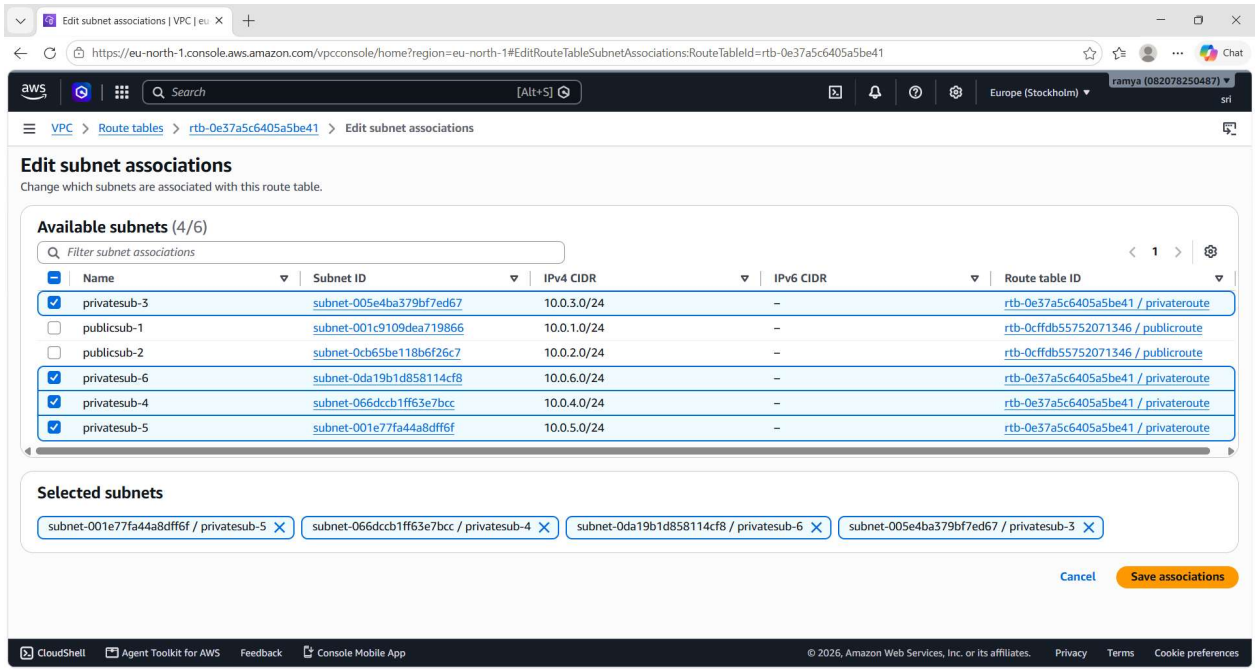


4. Create the Nat Gateway

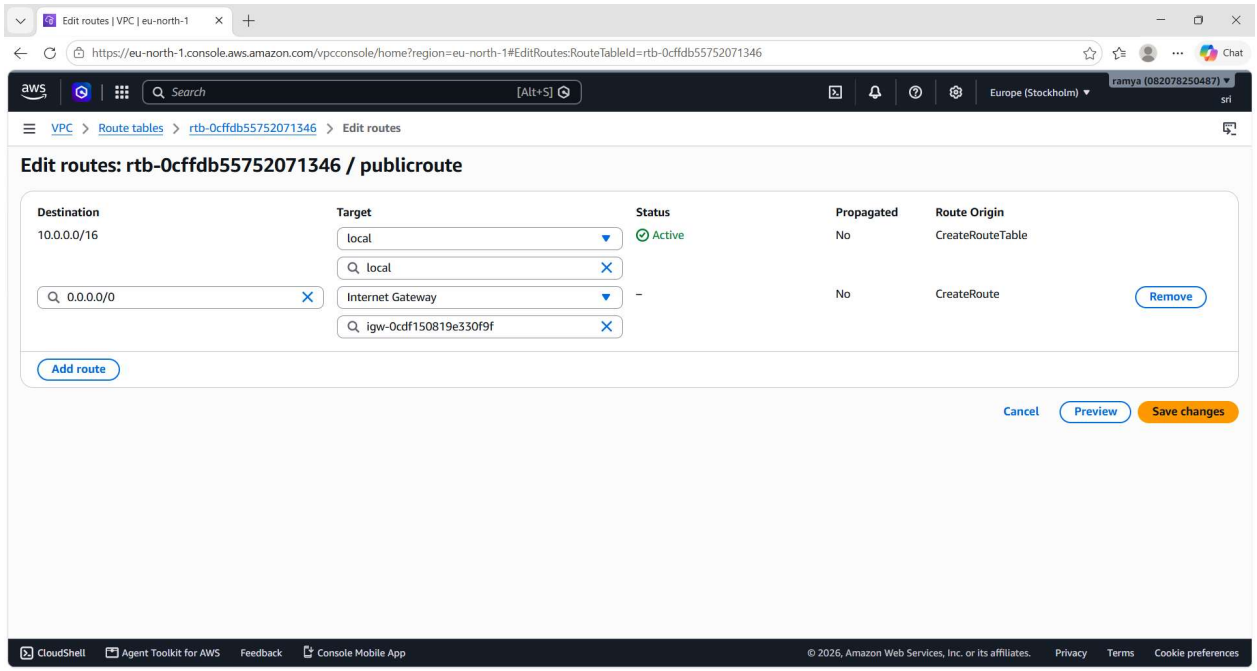


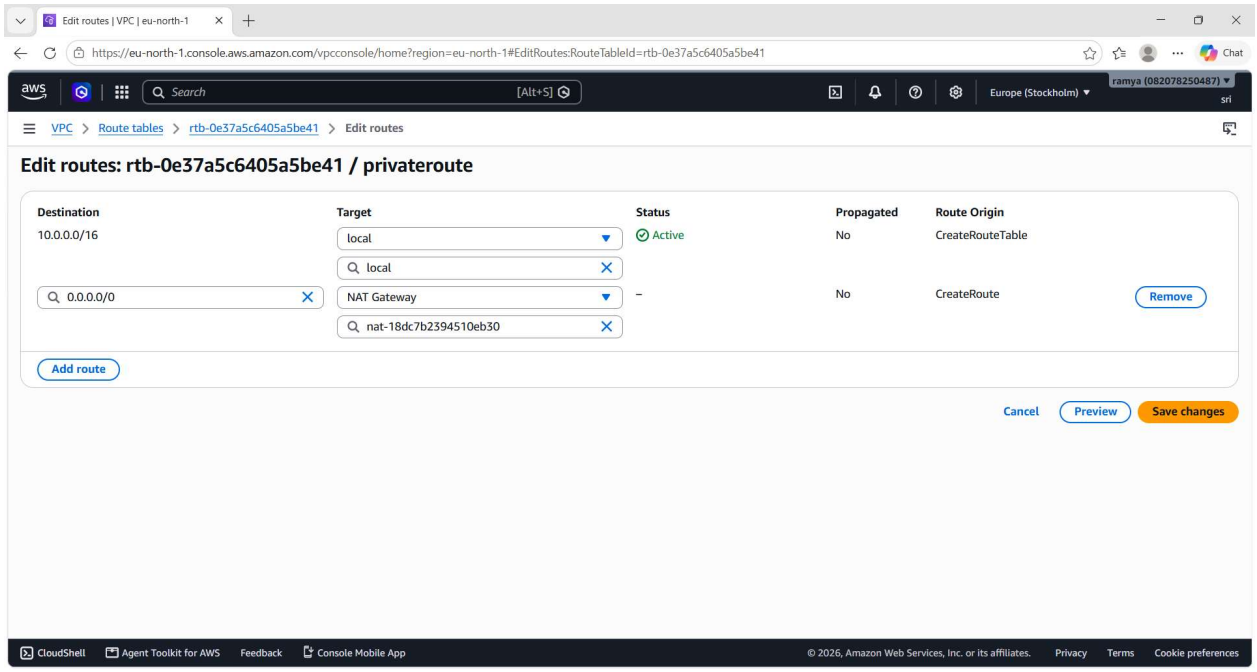
5. Create the route tables and associate the subnets with routetables



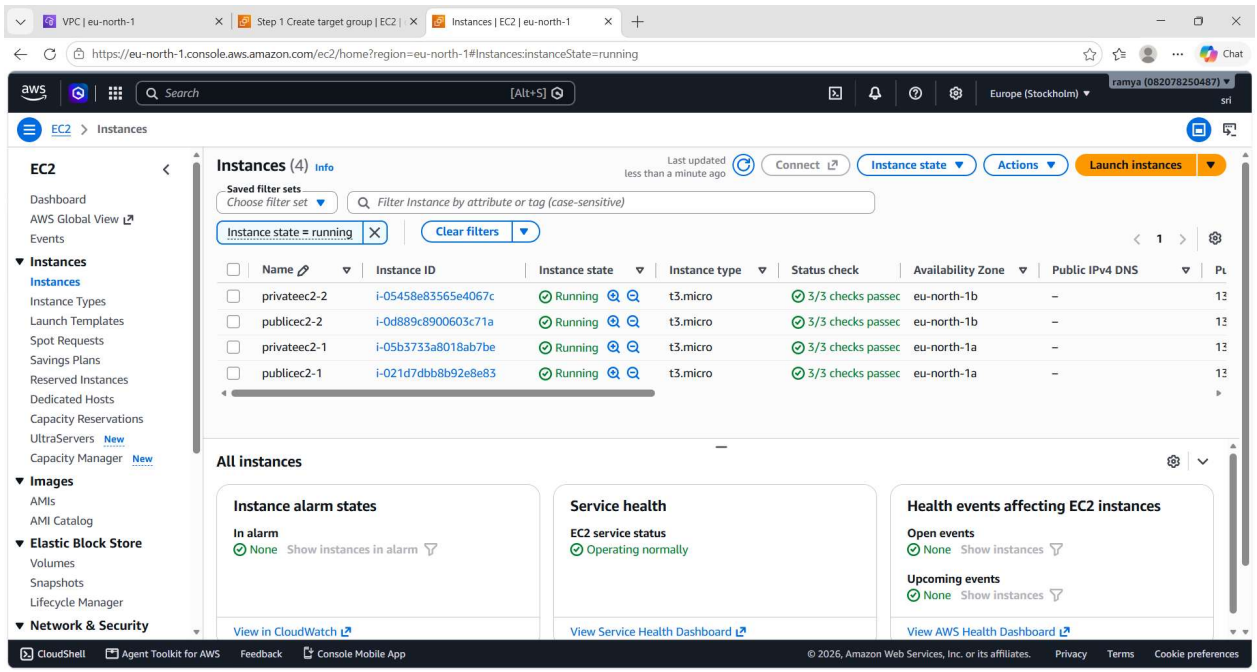


6. Attach the public routetable with internet gateway
   Attach the private routetable with NAT gateway



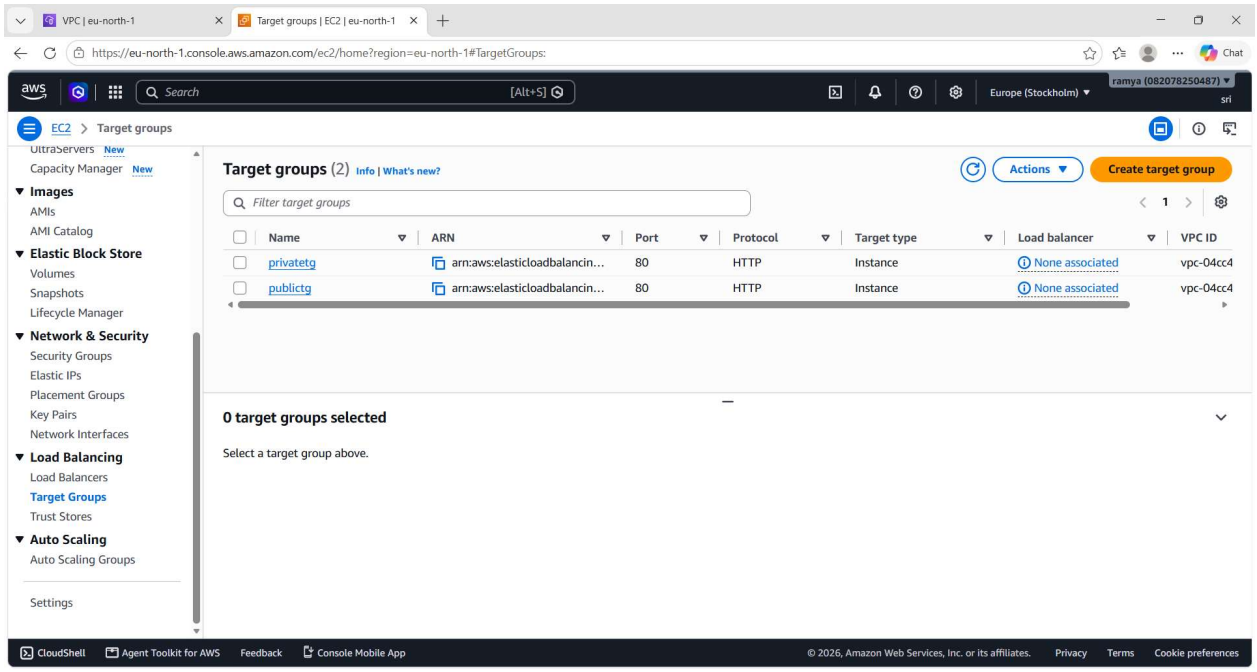


Step-2: Launch the EC2 instances - 2 public & 2 private

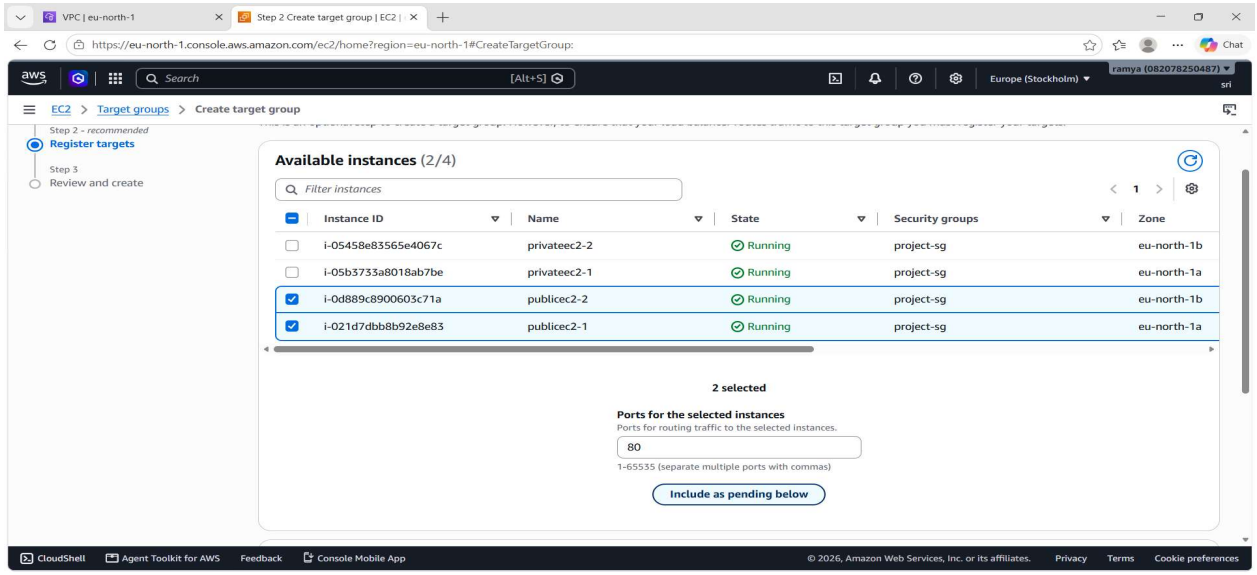


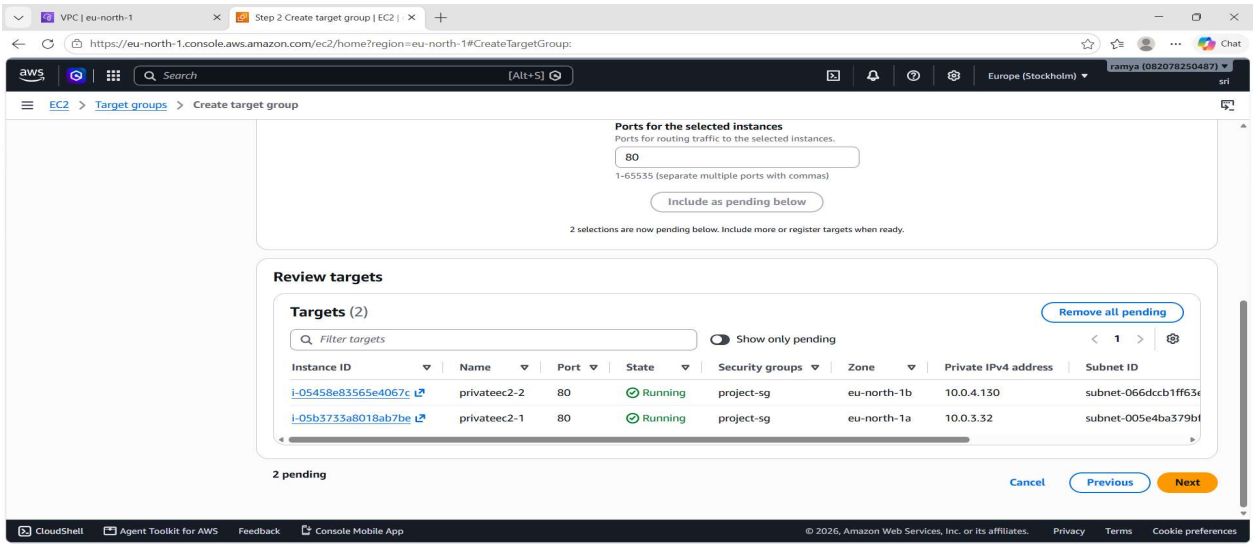
Step- 3: create the Loadbalancer

1.Create the target groups - public targetgrp & private targetgrp

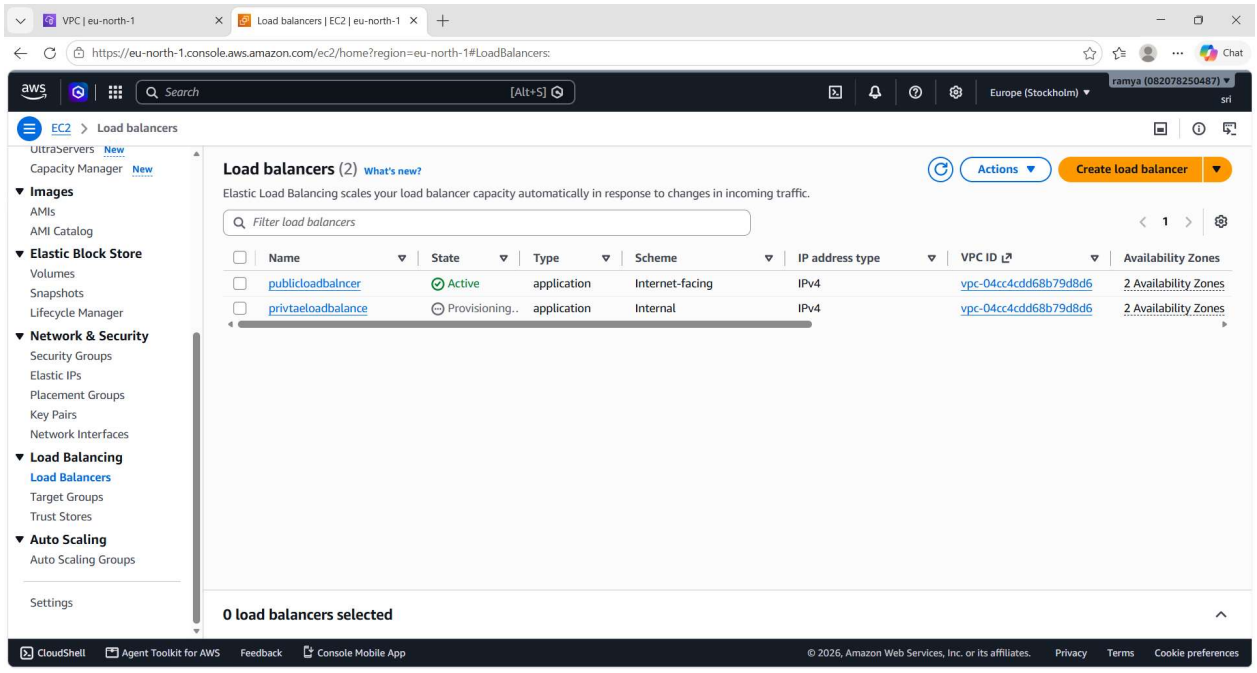


2. Associate the ec2 instance with their target groups

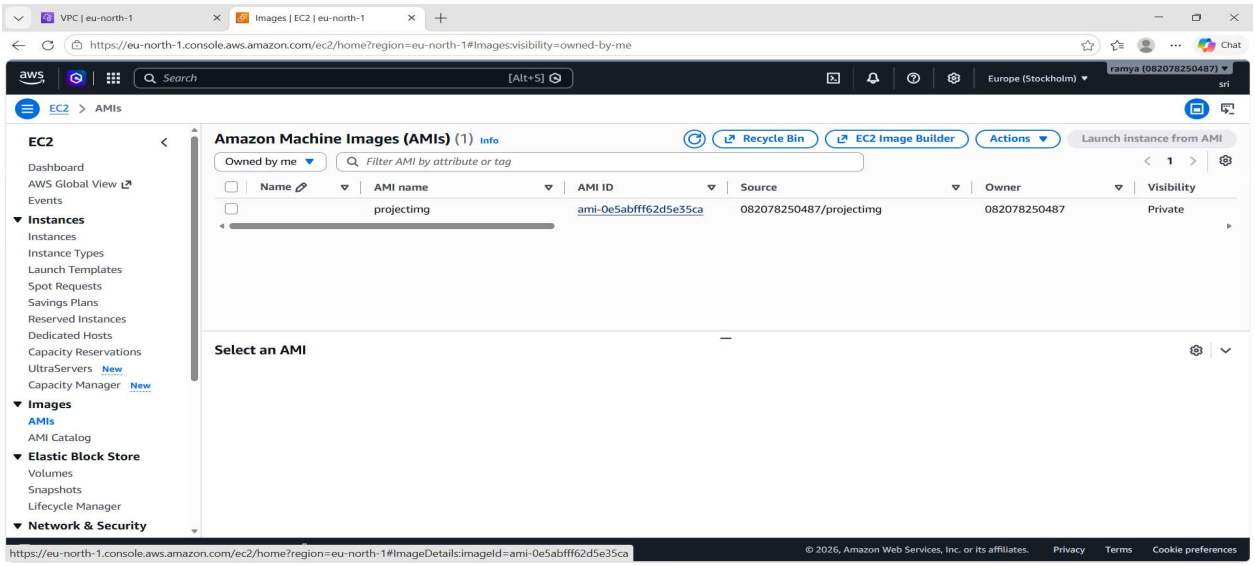




3. Create the loadbalancer
   - Public load balancer - Internet facing
   - Private load balancer- internal facing

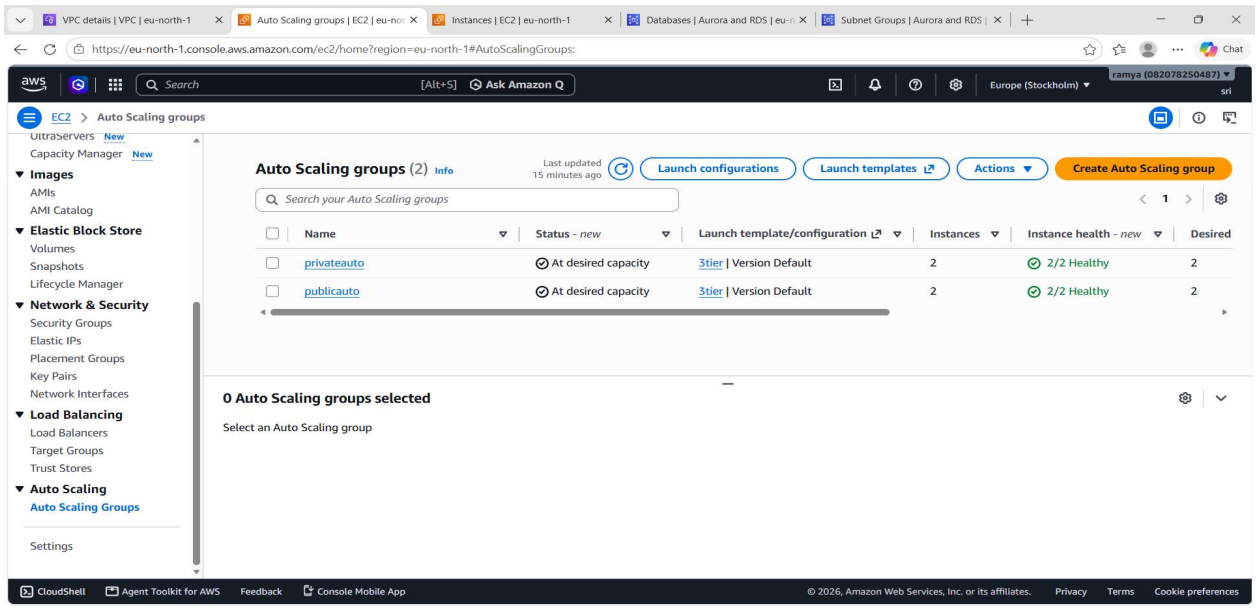


Step - 4 : Create the AMI(Amazon machine Image)

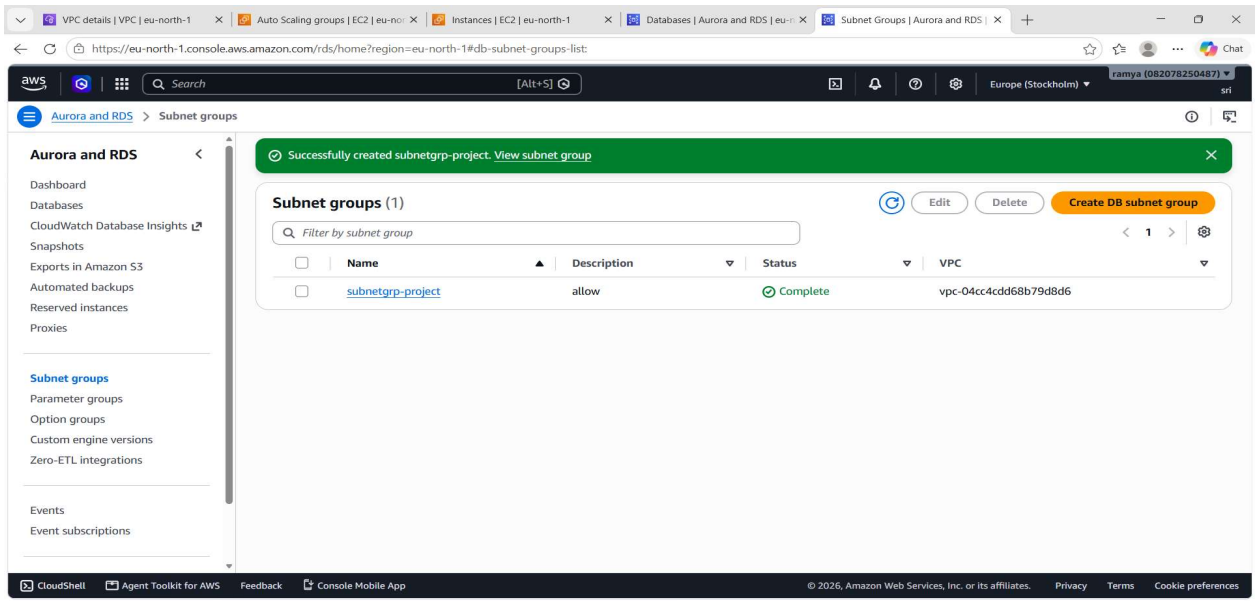


Step - 5: create the auto scaling grp

1. Create the launch template
2. Create the public and private auto scaling grp with desired capacity - 2 instances
3. It launches the extra 4 instances when the auto scaled grp reached the desired capacity

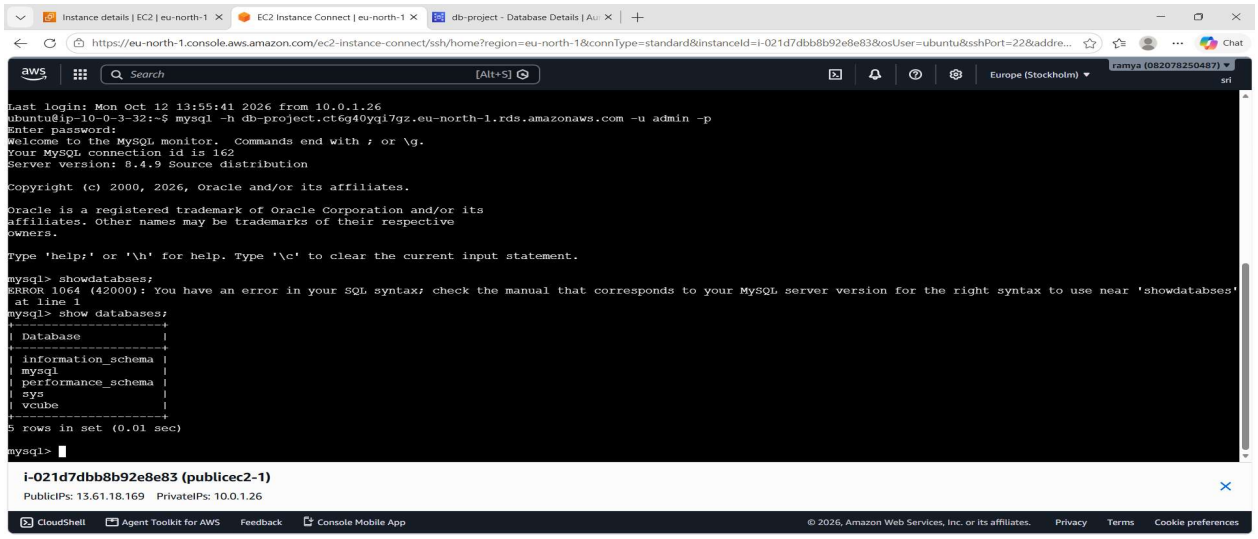


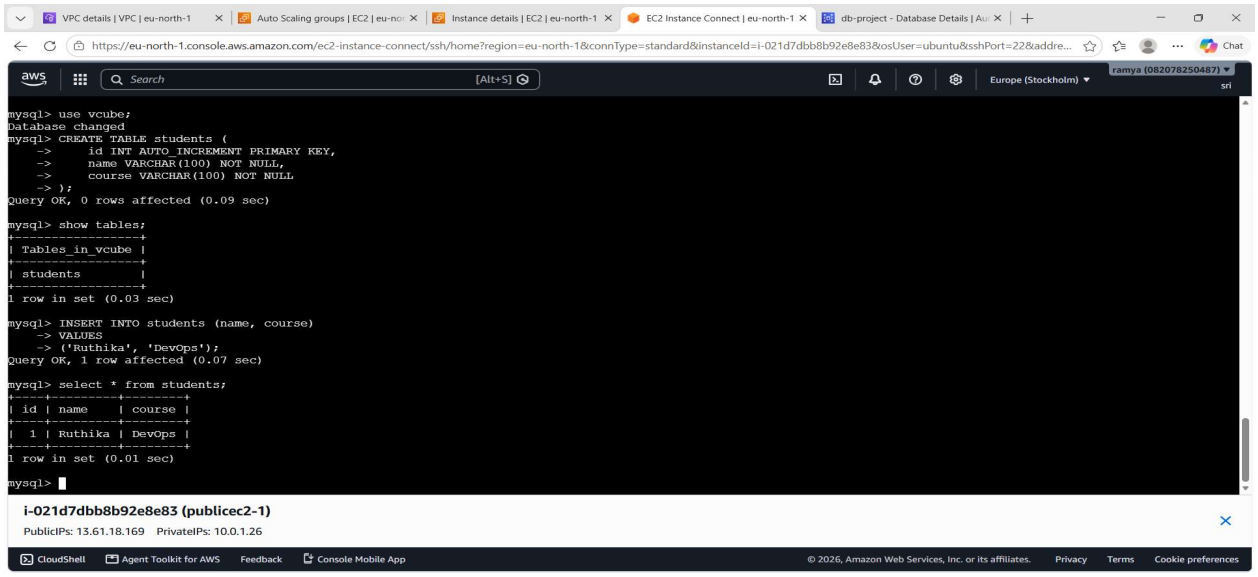
Step - 6: Create the DB subnet group



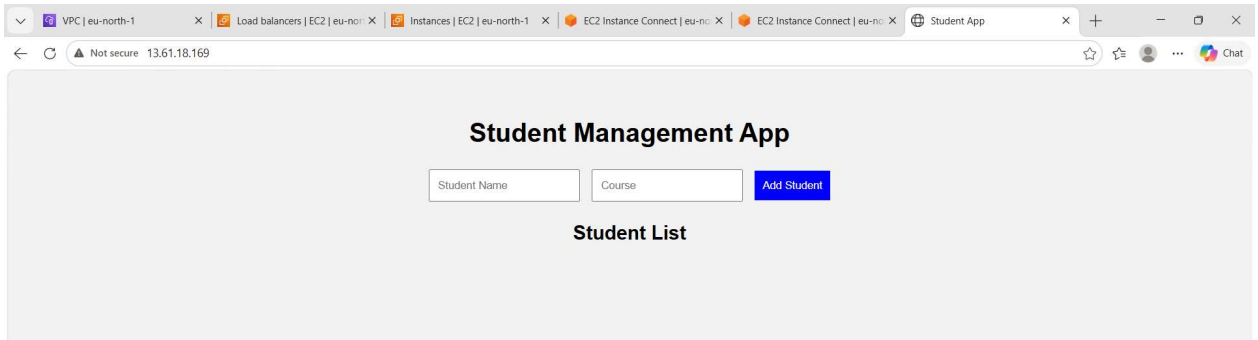
Step - 7: Create the Database

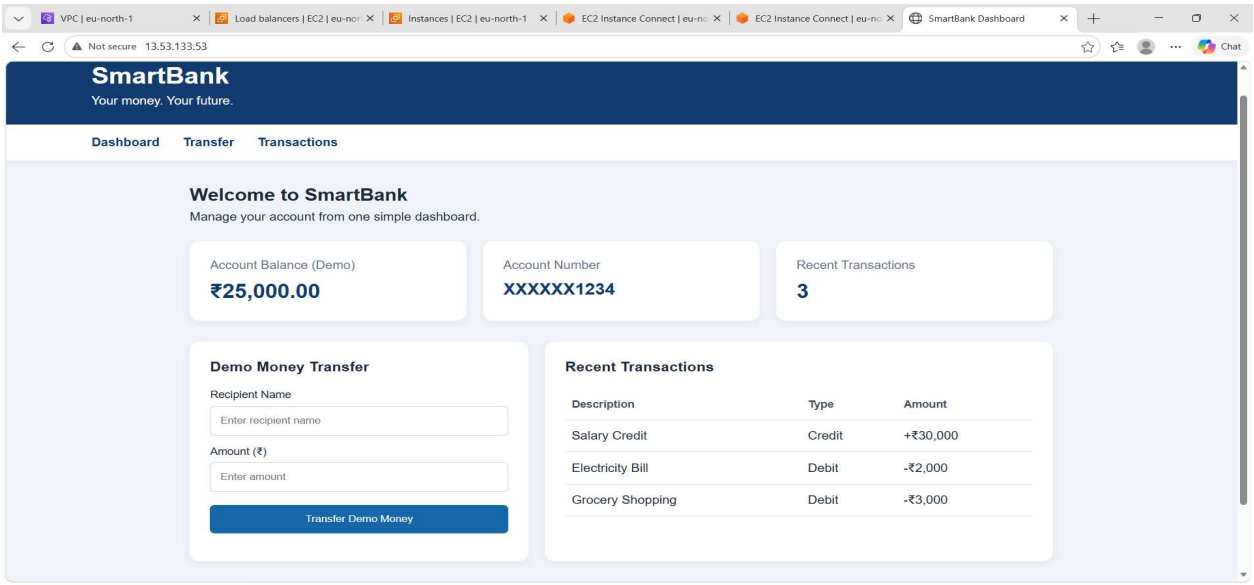
- This database is connected the private subnets
- Connect with public ec2 instance then private ec2 instance with PEM file.
- Install the mysql - sudo -i , apt update -y, sudo apt install mysql-server -y
- through endpoint with user,password and create the database and add tables.



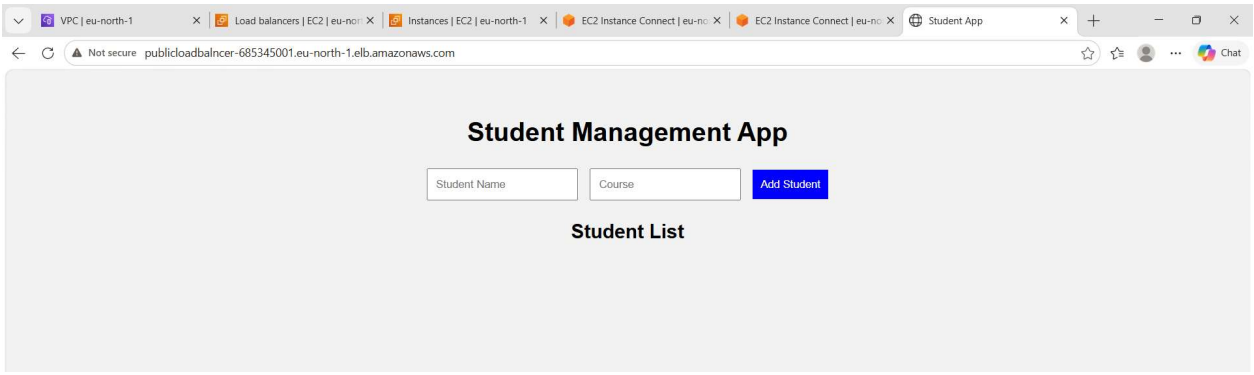


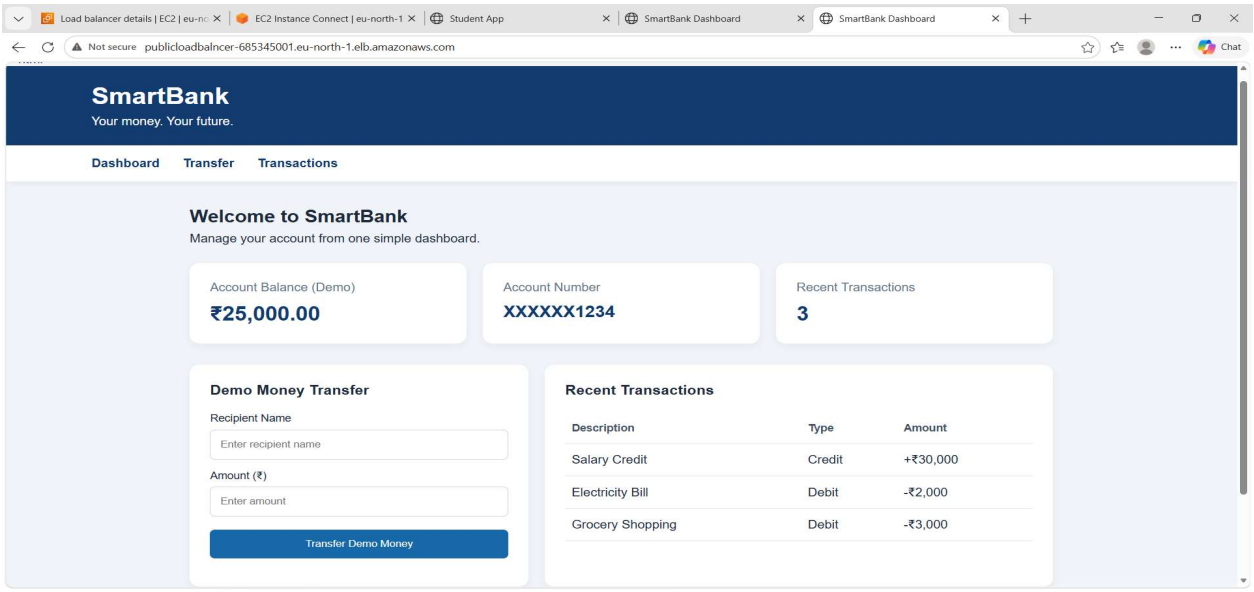
Step - 8 : Output through ip address





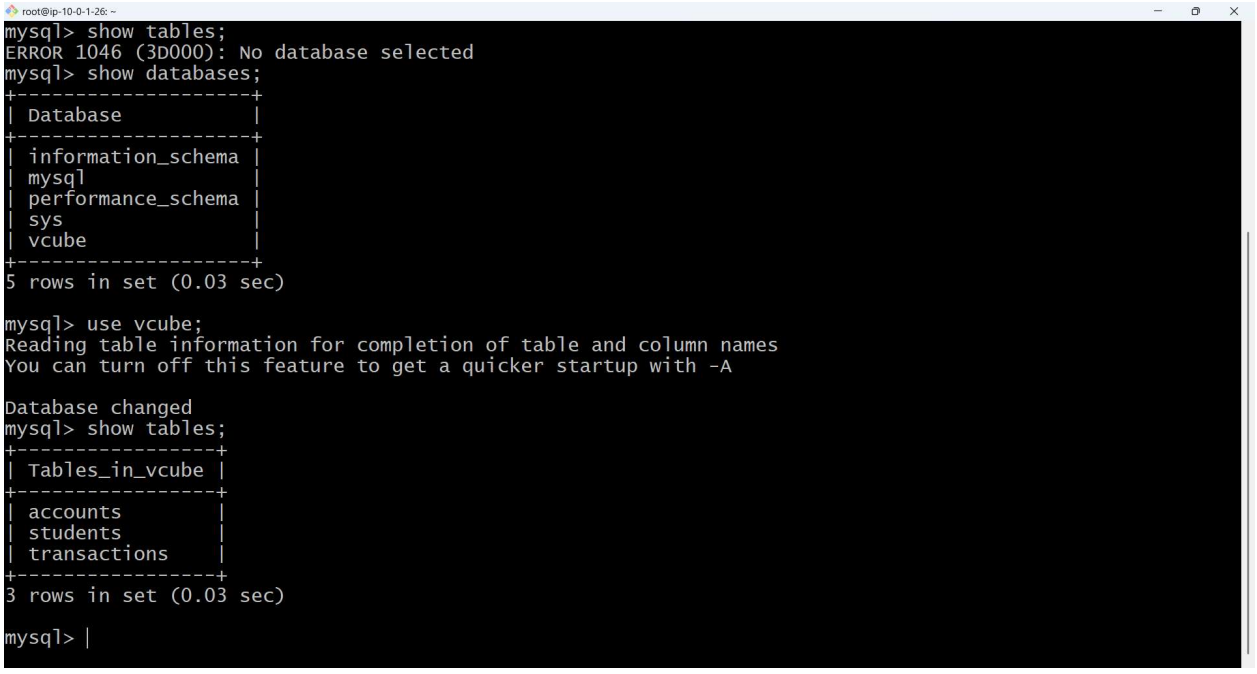
Step - 9: Output of ALB

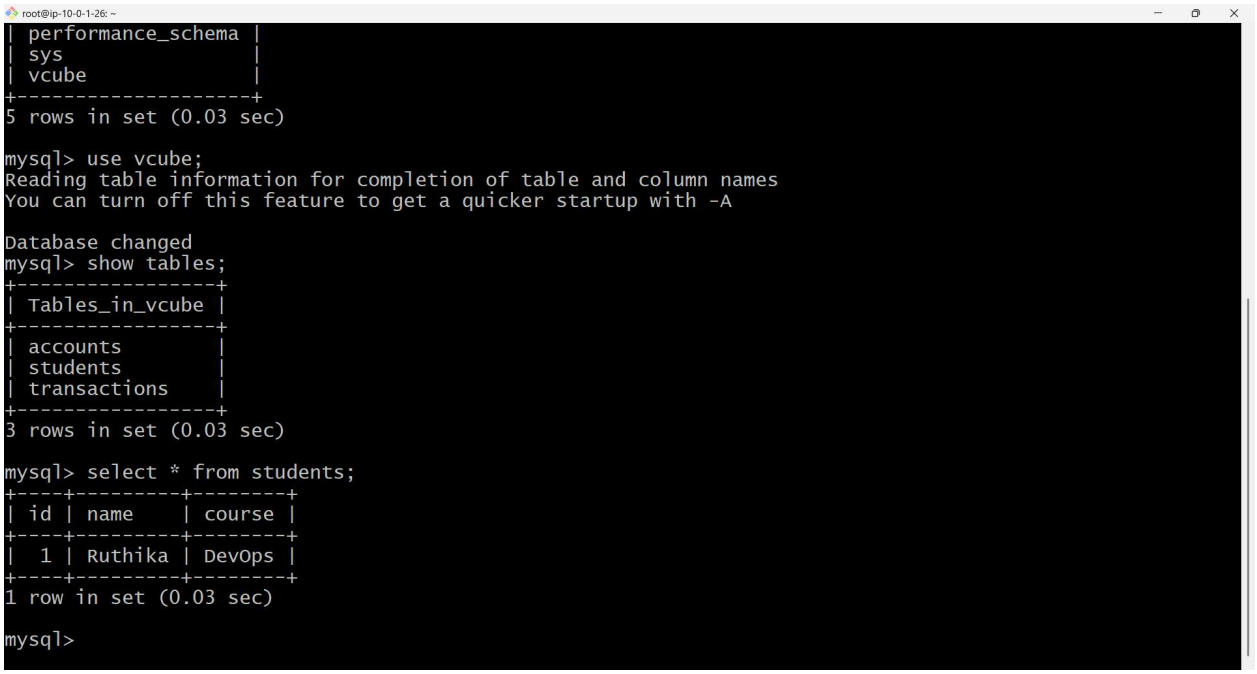


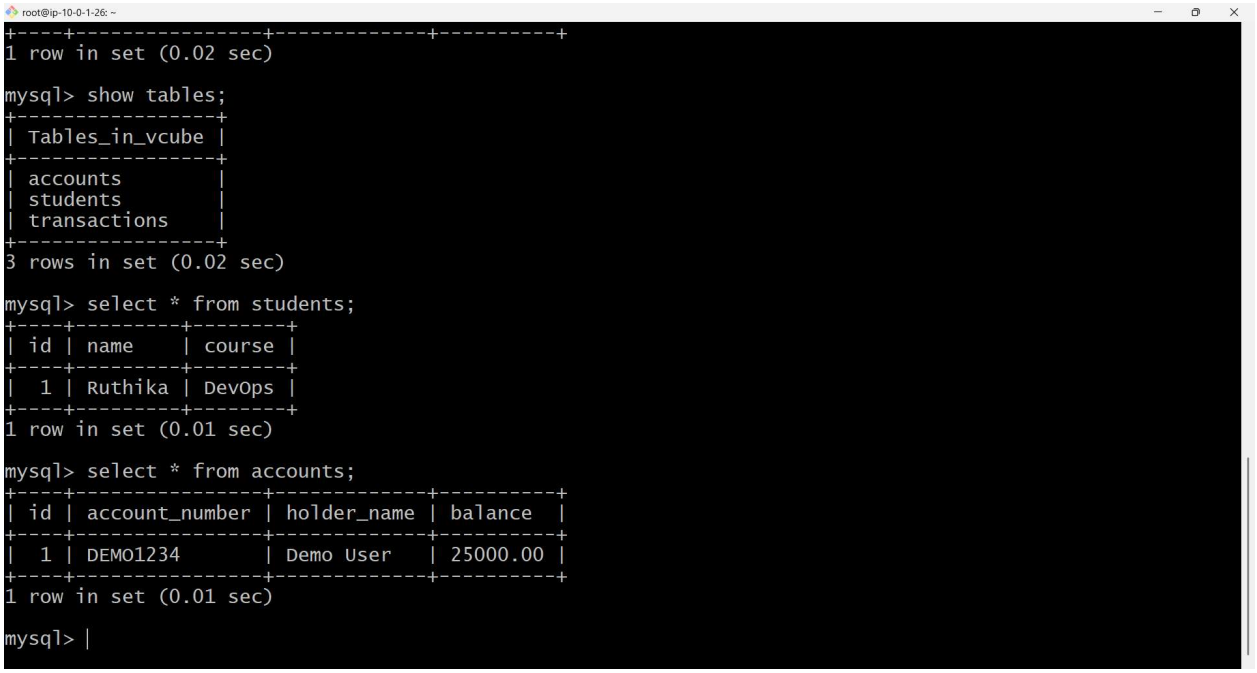


Step - 10: Output for database

- The database which we created in private instances
- Now we check in public ip address and connect with mysql




root@ip-10-0-1-26: ~
| performance_schema |
| sys |
| vcube |
+--------------------+
5 rows in set (0.03 sec)
mysql> use vcube;
Reading table information for completion of table and column names
You can turn off this feature to get a quicker startup with -A
Database changed
mysql> show tables;
+-----------------+
| Tables_in_vcube |
+-----------------+
| accounts |
| students |
| transactions |
+-----------------+
3 rows in set (0.03 sec)
mysql> select * from students;
+----+---------+--------+
| id | name | course |
+----+---------+--------+
| 1 | Ruthika | DevOps |
+----+---------+--------+
1 row in set (0.03 sec)
mysql>


root@ip-10-0-1-26: ~
+----+----------------+-------------+----------+
1 row in set (0.02 sec)
mysql> show tables;
+-----------------+
| Tables_in_vcube |
+-----------------+
| accounts |
| students |
| transactions |
+-----------------+
3 rows in set (0.02 sec)
mysql> select * from students;
+----+---------+--------+
| id | name | course |
+----+---------+--------+
| 1 | Ruthika | DevOps |
+----+---------+--------+
1 row in set (0.01 sec)
mysql> select * from accounts;
+----+----------------+-------------+----------+
| id | account_number | holder_name | balance |
+----+----------------+-------------+----------+
| 1 | DEMO1234 | Demo User | 25000.00 |
+----+----------------+-------------+----------+
1 row in set (0.01 sec)
mysql>